The rank-four, eleven-element excluded minors for representability over $\mathbb{F}_5$

A computational classification
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Abstract

There are exactly 35 isomorphism classes of rank-four, eleven-element excluded minors for representability over $\mathbb{F}_5$. The proof is an exhaustive construction. Every such matroid has a representable ten-element deletion. For each possible deletion, we recover all nonrepresentable extensions whose other deletions are representable, then test their contractions. The essential observation is that, when the ground set has more than twice the rank many elements, compatible basis data for the single-element deletions already determine a matroid. This turns the extension search into an exact intersection of finite tables. We prove that each stage preserves every possible excluded minor, explain how the tables are constructed and exhausted, and give the computational results that complete the classification. Separate certificates verify nonrepresentability and all immediate minors of each of the 35 resulting matroids.

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1 The theorem and the structure of its proof

A matroid is *quinary* if it is representable over $\mathbb{F}_5 = \mathbb{Z}/5\mathbb{Z}$. An *excluded minor* for quinary matroids is a nonquinary matroid every proper minor of which is quinary. We classify these obstructions at one specified rank and size.

Theorem 1.1 (Computational classification)**.** *There are exactly 35 isomorphism classes of rank-four, eleven-element excluded minors for representability over $\mathbb{F}_5$. Representatives $M_1, \ldots, M_{35}$ are specified in Appendix B.*

The defining condition has a useful finite form. A matroid M is an excluded minor precisely when

$$M \text{ is nonquinary}, \qquad M \setminus a \text{ and } M/a \text{ are quinary for every } a \in E(M).$$

Indeed, every proper minor is a minor of one of these immediate deletions or contractions, and field representability is closed under taking minors. For an eleven-element matroid there are therefore 22 immediate minors to consider. This criterion is easy to apply to a given matroid. The main difficulty is to find a finite list that is guaranteed to contain *every* matroid satisfying it.

We construct such a list by reversing deletion. Choose an element e of an unknown excluded minor M and put $N = M \setminus e$. The matroid N is quinary. We call it a *parent*, since the task is to recover M by adding e to it. The proof proceeds in three steps.

1. *Find all possible parents.* We prove that N must be a connected simple rank-four matroid on ten elements, and generate a complete quinary catalogue of these matroids (Section 2).

2. *Recover all possible extensions of each parent.* For a fixed N, we construct exactly the nonquinary extensions M for which every deletion $M \setminus a$ is quinary. The recovery theorem is proved in Section 3; its finite tables and their exhaustive solution are developed in Section 4.

3. *Impose the remaining minor conditions.* We test every contraction of every recovered matroid and identify isomorphic outputs. This leaves precisely the desired excluded minors (Section 5).

The second step is the heart of the argument. A quinary extension of N can be obtained by appending a column to a representation of N; the desired M, however, is nonquinary. Instead we construct its *different deletions* from representations, and require their basis data to agree wherever they overlap. A local-to-global lemma then shows that these compatible data form a matroid. We can thus enumerate nonrepresentable matroids using only finite lists of representable ones.

Two inclusions will finish the proof. The construction must contain every excluded minor, giving an upper bound on their number. Each listed object must also be an excluded minor, and the objects must be pairwise nonisomorphic, giving the matching lower bound. The computation in Section 6 supplies both: the exhaustive construction leaves 35 classes, and separate certificates verify all 35. Its numerical results are organized there according to the three steps above; implementation records are collected in Appendix C.

For comparison, Brettell's classification through ten elements gives 5, 92, 219, and 130 rank-four excluded minors at sizes seven through ten [1]. We reproduce those smaller catalogues as checks on the method. The use of compatible extensions and of inequivalent representations has precedents in the computations of Brettell and Pendavingh [2]. Here the proof of coverage rests on generating all possible parents and imposing compatibility with every deletion.

2 Step 1: finding every possible parent

This step establishes the starting point of the search: a complete list of matroids N that could occur as $M \setminus e$. We first determine which parents are needed, then prove that the represented generation procedure reaches all of them.

2.1 Why connected simple parents suffice

We use the standard notation $r_M(X)$ for the rank of a subset and $\mathcal{B}(M)$ for the bases of M. All matroids are finite. A simple matroid has no loops or parallel pairs.

Lemma 2.1 (Possible deletions)**.** *If M is a rank-four, eleven-element excluded minor for quinary matroids, then for every $e \in E(M)$ the deletion $M \setminus e$ is a connected simple rank-four quinary matroid on ten elements.*

Proof. An excluded minor for representability over a field is 3-connected when it has at least four elements. To recall the reason, representability over a fixed field is closed under direct sums and 2-sums. A disconnected matroid decomposes as a direct sum; a connected matroid with a 2-separation decomposes as a 2-sum. The summands are isomorphic to proper minors [3]. If the matroid were an excluded minor, all summands would be representable, forcing the matroid itself to be representable. For the 2-sum construction, choose representations in which the common element is a unit vector, identify its one-dimensional span in the two vector spaces, and delete its column.

Thus M is 3-connected. It has no loops, parallel pairs, or coloops, so $N = M \setminus e$ is simple and has rank four. It is quinary by minor-minimality. It remains to prove that N is connected. If $E(N) = A \dot\cup B$ were a nontrivial direct-sum partition, choose the names so that $|A| \geq 2$. Since $r(N) = r(M)$, we would have

$$r_M(A) + r_M(B) = r(M), \qquad r_M(A) + r_M(B \cup \{e\}) - r(M) \leq 1.$$

Both A and $B \cup \{e\}$ have at least two elements. This contradicts 3-connectivity of M. □

We call a matroid *labelled* when its element names are fixed and its bases are specified as subsets of that named ground set. Let $P = \{0, \ldots, 9\}$ and take a new label $e = 10$. Let $\mathcal{P}$ contain one labelled representative on P of each connected simple rank-four quinary matroid on ten elements. For each $N \in \mathcal{P}$, define $\mathcal{D}(N)$ to be the set of simple rank-four matroids M on $P \cup \{e\}$ satisfying

$$M \setminus e = N, \qquad M \text{ is nonquinary},$$
$$M \setminus a \text{ is quinary for every } a \in P \cup \{e\}.$$

We call these *deletion-minimal nonquinary extensions* of N. The word "deletion" is essential: contractions have not yet been tested.

Proposition 2.2 (Reduction of the classification)**.** *The rank-four, eleven-element quinary excluded minors are exactly, up to isomorphism, the members of*

$$\bigcup_{N \in \mathcal{P}} \{M \in \mathcal{D}(N) : M/a \text{ is quinary for every } a \in E(M)\}.$$

Proof. Every matroid in the displayed set is nonquinary and has quinary immediate deletions and contractions, so it is an excluded minor. Conversely, let M be any excluded minor of the specified rank and size. By Lemma 2.1, a deletion $M \setminus e$ is isomorphic to a member N of $\mathcal{P}$. Relabel this deletion as N and its missing element as e. The resulting matroid belongs to $\mathcal{D}(N)$, and all its contractions are quinary. Its isomorphism class therefore occurs in the displayed set. □

This proposition specifies exactly what the rest of the computation must produce. It suffices to construct $\mathcal{P}$, compute each $\mathcal{D}(N)$ completely, and apply the contraction test. In particular, a parent may be discarded for being disconnected, but not merely for failing to be 3-connected.

2.2 Generating all simple quinary parents

The aim is to list, up to matroid isomorphism, all simple rank-four quinary matroids on ten elements, and then retain the connected ones. We describe the construction for a general rank r and target size m; here $r = 4$ and $m = 10$. Throughout this construction, k denotes the current number of elements, ranging from r to m.

A rank-r quinary matroid on k elements has a full-rank $r \times k$ representation over $\mathbb{F}_5$, with one column per element. In a *labelled representation*, the column indexed by u represents the element with label u. Simplicity means that no column is zero and no two columns are scalar multiples. Thus its columns specify distinct points of $\mathrm{PG}(r-1, 5)$. We represent each projective point by the vector whose first nonzero entry is 1. There are $(5^r - 1)/4$ such points, in particular 156 in rank four.

Two matrices are *field-isomorphic* if one can be obtained from the other by invertible row operations, nonzero column scalings, and a permutation of columns. Field-isomorphic matrices represent isomorphic matroids (but the converse can fail). During growth we retain one matrix from every field-isomorphism class.

Every full-rank representation contains r columns forming a basis. Move them to the front and apply row operations to make them I_r. This explains both the starting matrix and the growth procedure:

1. Start at size $k = r$ with the single matrix I_r.
2. From every retained $r \times k$ matrix, append each projective point not already among its columns. These are all its simple represented extensions of rank r and size $k + 1$.
3. Among all matrices produced at size $k + 1$, retain one per field-isomorphism class. Repeat until size m is complete.

The smaller levels $r, \ldots, m-1$ are the intermediate stages used to construct the target level, rather than enumerating all sets of m projective points directly.

The implementation tests field isomorphism using Sage's `is_field_isomorphic`.

Proposition 2.3 (Completeness of represented generation)**.** *For every $k = r, \ldots, m$, the procedure retains a matrix from every field-isomorphism class of simple rank-r quinary representations on k elements. Consequently, their underlying matroids include every simple rank-r, k-element quinary matroid up to isomorphism.*

Proof. Fix a simple rank-r representation A on $s \le m$ elements. Choose a basis, move its columns to the front, and transform them to I_r. Order the remaining columns, and let A_k be the first k columns, for $r \le k \le s$. Thus $A_r = I_r$ and $A_s = A$.

We prove by induction on k that the retained level at size k contains a matrix field-isomorphic to A_k. At size r this is the starting matrix I_r. Suppose the claim holds at size $k < s$, and let B_k be a retained matrix field-isomorphic to A_k. Write $A_{k+1} = [A_k \mid v]$. Let T be the invertible row transformation in an equivalence taking A_k to B_k, the other operations being scalings and a permutation of its old columns. Then

$$[B_k \mid Tv] \quad \text{is field-isomorphic to} \quad [A_k \mid v].$$

Indeed, apply the same row transformation to the extended matrix, and extend the old column operations by leaving the new column fixed. Normalize Tv to represent its projective point. This point is unused in B_k: otherwise invertibility of T would imply that v is parallel to an old column of A_k, contrary to simplicity.

The procedure tries every unused projective point, so it generates this extension of B_k. Deduplication retains either that matrix or a field-isomorphic one. This proves the induction step and hence the claim at size s. Since A and s were arbitrary, the assertion follows. □

To obtain the abstract catalogue from a completed represented level, replace each matrix A by its basis family

$$\mathcal{B}(A) = \{S \subseteq \{0, \dots, k-1\} : |S| = r,\ \det(A_S) \neq 0\},$$

where A_S consists of the columns indexed by S. A basis family specifies the whole matroid without using coordinates. Two such families define isomorphic matroids exactly when a permutation of the elements takes one family to the other. Sage's exact `is_isomorphic` test on basis matroids removes these abstract repetitions. Each retained abstract matroid is stored as one bit for each r-subset in lexicographic order, recording whether it is a basis; for a ten-element rank-four parent this uses $\binom{10}{4} = 210$ bits, stored in hexadecimal.

The matrix catalogue is saved separately and remains the input for further represented growth and the labelled representation lists in Section 4. At size ten, we retain the connected abstract rank-four matroids as $\mathcal{P}$. The same generation procedure in rank three supplies the catalogue used to test contractions in Section 5.

Step 1 is now justified: every possible deletion of an excluded minor occurs in $\mathcal{P}$. We next explain how to recover its possible nonquinary extensions.

3 Step 2: recovering an extension from its deletions

Fix one parent $N \in \mathcal{P}$. Our goal in this section is an exact description of $\mathcal{D}(N)$ that can be computed from finite lists of quinary matroids. We first describe the unknown data of an extension, then express the conditions on its deletions in those data. The main proposition will show that these conditions already enforce the matroid axioms.

3.1 The unknown basis data

The deletion equation $M \setminus e = N$ fixes the status of every four-set avoiding e: it is a basis of M exactly when it is a basis of N. The only unknown four-sets are therefore

$$T \cup \{e\}, \qquad T \in \binom{P}{3}.$$

There are $\binom{10}{3} = 120$ of these, compared with the $\binom{10}{4} = 210$ already fixed four-sets. Introduce a Boolean variable x_T for each triple, with $x_T = 1$ meaning that $T \cup \{e\}$ is a basis. An assignment x specifies a nonempty four-subset family

$$\mathcal{B}_N(x) = \mathcal{B}(N)\ \cup \{T \cup \{e\} : T \in \binom{P}{3},\ x_T = 1\}.$$

For the moment this is only a proposed basis family; an arbitrary assignment need not define a matroid.

Consider how another deletion reads these variables. For $f \in P$, the four-sets of $M \setminus f$ that contain e are indexed by the triples avoiding f. Write

$$\pi_f(x) = x\big|_{\binom{P\setminus\{f\}}{3}}$$

for the restriction to those coordinates. The four-sets avoiding both e and f are already determined by $N \setminus f$. Thus $\pi_f(x)$ is exactly the extra basis data needed to specify $M \setminus f$ as an extension of $N \setminus f$.

For example, the bit $x_{\{0,1,2\}}$ records whether $\{0, 1, 2, 10\}$ is a basis. This bit is visible in the deletion of 3 and in the deletion of 4, but disappears in the deletion of 0. If independently chosen extensions for the deletions of 3 and 4 disagree on that bit, they cannot be deletions of one matroid. A single assignment x enforces agreement on every such shared four-set at once.

To express which deletions are permitted, let Q be any fixed labelled simple rank-four quinary matroid on a set U. Define its *representable extension table* $\mathcal{R}(Q)$ to be the set of Boolean patterns z on $\binom{U}{3}$ for which

$$\mathcal{B}(Q) \ \cup \{T \cup \{e\} : z_T = 1\}$$

is the basis family of a simple rank-four extension of Q that admits a representation over $\mathbb{F}_5$. We construct this table from representations of Q in Section 4.

The meaning of table membership is now precise:

- $\pi_f(x) \in \mathcal{R}(N \setminus f)$ says that the proposed deletion avoiding f is simple, rank four, and quinary;
- $x \in \mathcal{R}(N)$ says that the proposed full extension of N is itself simple, rank four, and quinary.

The first statement is meaningful because a connected parent has no coloops, so every $N \setminus f$ still has rank four.

Consequently the assignments we want should form the finite set

$$\mathcal{S}(N) = \left\{x \in \{0,1\}^{\binom{P}{3}} : \pi_f(x) \in \mathcal{R}(N \setminus f) \text{ for every } f \in P\right\} \setminus \mathcal{R}(N).$$

The intersection requires all the other deletions to be quinary; the subtraction removes the extensions that are quinary themselves. What still needs proof is that an assignment satisfying these conditions actually defines a matroid. Without that implication, the displayed expression would merely describe consistent tables of four-subsets.

3.2 Why compatible deletions force the matroid axioms

The basis-exchange axiom involves two bases and hence at most twice the rank many elements. If the ground set is larger, the whole exchange question can be seen inside one deletion. This gives the needed local-to-global principle.

Lemma 3.1 (Deletion compatibility)**.** *Let $|E| = n > 2r$ and let $\varnothing \neq \mathcal{B} \subseteq \binom{E}{r}$. Suppose that for each $a \in E$ the family of members of $\mathcal{B}$ avoiding a is the basis family of a rank-r matroid on $E \setminus \{a\}$. Then $\mathcal{B}$ is the basis family of a rank-r matroid on E.*

Proof. Take $A, B \in \mathcal{B}$ and $b \in A \setminus B$. We must find $c \in B \setminus A$ such that $(A \setminus \{b\}) \cup \{c\} \in \mathcal{B}$. Since $|A \cup B| \leq 2r < n$, there is an element $a \notin A \cup B$. Both A and B are bases of the specified deletion avoiding a. Basis exchange in that matroid supplies the required c, and its exchanged basis also belongs to $\mathcal{B}$. This proves every ordered instance of basis exchange. Together with nonemptiness and the common size of the members, it proves that $\mathcal{B}$ is a matroid basis family. □

Here $n = 11$ and $r = 4$, so the lemma applies. In particular, the representations witnessing different deletions do not have to fit together as matrices. They need only prescribe compatible bases. The lemma then supplies the abstract matroid, even when it has no representation over $\mathbb{F}_5$.

Proposition 3.2 (Exact recovery of deletion-minimal extensions)**.** *For a fixed parent $N \in \mathcal{P}$, the map $x \mapsto M_x$, where $\mathcal{B}(M_x) = \mathcal{B}_N(x)$, is a bijection from $\mathcal{S}(N)$ to $\mathcal{D}(N)$.*

Proof. First take $x \in \mathcal{S}(N)$. The proposed basis family $\mathcal{B}_N(x)$ is nonempty because it contains $\mathcal{B}(N)$. Its deletion avoiding e is the rank-four quinary matroid N. For each $f \in P$, the condition $\pi_f(x) \in \mathcal{R}(N \setminus f)$ says that its deletion avoiding f is also the basis family of a simple rank-four quinary matroid. Lemma 3.1 therefore makes $\mathcal{B}_N(x)$ a matroid basis family.

Call this matroid M_x. Each pair of elements lies in one of its simple deletions, so M_x is simple. All its deletions are quinary by construction. If M_x were quinary, its pattern would belong to $\mathcal{R}(N)$ by the definition of that table, contrary to the subtraction in $\mathcal{S}(N)$. Hence $M_x \in \mathcal{D}(N)$.

Conversely, take $M \in \mathcal{D}(N)$. Its bases containing e give a unique pattern x. For every $f \in P$, the matroid $M \setminus f$ is a simple quinary extension of $N \setminus f$; it has rank four since it contains the rank-four restriction $N \setminus f$. Thus $\pi_f(x) \in \mathcal{R}(N \setminus f)$. Since M is nonquinary, $x \notin \mathcal{R}(N)$. Therefore $x \in \mathcal{S}(N)$, and $M = M_x$. Finally, different patterns give different basis families, proving injectivity. □

This proposition is the bridge between the mathematical problem and the computation. It accounts for all four requirements on an output: validity as a matroid, simplicity, representability of every deletion, and nonrepresentability of the full matroid. To finish Step 2, it remains to build the tables $\mathcal{R}(Q)$ without omissions and exhaust the finite conditions defining $\mathcal{S}(N)$.

4 Constructing and exhausting the extension tables

For each $Q = N$ or $Q = N \setminus f$, we construct $\mathcal{R}(Q)$ as a Boolean table with $\binom{|U|}{3}$ columns, one per triple of $U = E(Q)$. Write each triple in increasing label order and sort the triples lexicographically. Each row describes one representable extension of Q by e. Its entry in column T is 1 exactly when $T \cup \{e\}$ is a basis.

Each deletion inherits its labels from P. For example, the column for $\{1, 2, 3\}$ is first in $\mathcal{R}(N \setminus 0)$ and occurs later in $\mathcal{R}(N \setminus 4)$. Both entries record the status of the same four-set $\{1, 2, 3, e\}$, so the compatibility check compares entries by triple label.

We first collect the labelled representations $\mathcal{L}(Q)$ and use them to generate the rows of $\mathcal{R}(Q)$. We then compress the columns of these tables and search for full rows whose restrictions occur in every deletion table.

4.1 Complete labelled representation sets

Two labelled representations of Q are *labelled projectively equivalent* if invertible row operations and nonzero column scalings take one to the other, with each column retaining its label. We seek a set $\mathcal{L}(Q)$ containing exactly one matrix from each of these equivalence classes.

Lemma 4.1 (Complete labelled representations)**.** *For each field-isomorphism class in the catalogue representing a matroid isomorphic to Q, choose one matrix and relabel its columns using one matroid isomorphism to Q. Apply every automorphism of Q to the column labels of these matrices. The resulting set contains every labelled projective equivalence class of Q.*

Proof. Take any labelled representation of Q. Its field-isomorphism class occurs in the catalogue. A field isomorphism from the chosen catalogue matrix to this representation gives a bijection of its column labels with U. This bijection and the chosen relabelling to Q are both matroid isomorphisms, so they differ by an automorphism of Q. Applying that automorphism therefore reaches the desired representation up to row operations and column scaling. □

To remove repetitions, fix an ordered basis $B = (b_1, \dots, b_r)$ of Q and use Gaussian elimination to make its columns I_r. Write Z for the remaining columns, keeping their labels. Its zero pattern is fixed by Q, since

$$Z_{ij} \neq 0 \quad \Longleftrightarrow \quad (B \setminus \{b_i\}) \cup \{j\} \in \mathcal{B}(Q), \qquad j \in U \setminus B.$$

Form the bipartite support graph with row vertices i, column vertices $j \in U \setminus B$, and edges at the nonzero entries of Z. Choose a fixed spanning forest of this graph. Row and column scalings can make every forest entry equal to 1; inverse scalings on the basis columns preserve I_r.

Lemma 4.2 (Normal form)**.** *With the ordered basis and support forest fixed, this normalization gives exactly one matrix per labelled projective equivalence class.*

Proof. For existence, choose a scale at one vertex of each tree and traverse its edges. Each edge uniquely determines the scale at its unvisited endpoint that makes its entry 1. A tree has no cycle, so these choices cannot conflict.

For uniqueness, two equivalent matrices reduced on the same labelled basis differ only by row and column scalings: the row transformation must be diagonal to preserve each basis-column direction. If both matrices have forest entries 1, each forest edge forces the product of its endpoint multipliers to be 1. Within each component, all row multipliers are therefore equal and all column multipliers are their inverses. Every other edge has endpoints in the same component, so its entry is unchanged too. Isolated vertices affect no entry of Z. Thus the normalized matrices agree. □

The calculation now reduces to matroid isomorphism, graph automorphism, and matrix normalization. Sage supplies a relabelling to Q for each catalogue matrix. To obtain $\mathrm{Aut}(Q)$, form the bipartite incidence graph of elements and bases (or nonbases), giving the two types of vertex different colors; its color-preserving automorphisms induce exactly $\mathrm{Aut}(Q)$ on U. Sage supplies generators of this group. Starting from the aligned matrices, repeatedly apply these generators, normalize each result, and retain it only if that normalized matrix is new. This finite orbit calculation yields $\mathcal{L}(Q)$ by the two lemmas above.

4.2 Building a table by appending projective points

For each $V \in \mathcal{L}(Q)$, append every normalized projective point $v \in \mathrm{PG}(3,5)$ whose direction is unused among the columns of V. The matrix $[V \mid v]$ represents a simple rank-four extension of Q with the appended column indexed by e.

Define its Boolean row $z(V, v)$ by

$$z(V,v)_T = 1 \quad \Longleftrightarrow \quad \det[\, V_T \mid v \,] \neq 0, \qquad T \in \binom{U}{3},$$

and set the bit to 0 otherwise. Here V_T consists of the three old columns indexed by T. Different matrices can give the same row. Collect the distinct Boolean rows in the triple order given above.

Proposition 4.3 (Completeness of the extension table)**.** *The rows obtained in this way are exactly $\mathcal{R}(Q)$.*

Proof. Every matrix $[V \mid v]$ gives a simple rank-four quinary extension, so its row belongs to $\mathcal{R}(Q)$. Conversely, let $z \in \mathcal{R}(Q)$. By definition of this table, the basis family

$$\mathcal{B}(Q) \ \cup \{T \cup \{e\} : z_T = 1\}$$

defines a simple rank-four quinary matroid K. Choose a matrix $[W \mid w]$ representing K, with old columns labelled by U and new column labelled e. The submatrix W represents Q, so it is labelled projectively equivalent to some $V \in \mathcal{L}(Q)$. Apply the row transformation and old-column scalings to the whole matrix, giving $[V \mid v]$. Normalize v by scaling its column. Since K is simple, its direction is nonzero and unused in V. The construction therefore tries this pair (V, v) and records exactly z. □

For a fixed V, these determinant tests reduce to one matrix product over $\mathbb{F}_5$. For each triple T, compute the signed cofactor vector $c_T \in \mathbb{F}_5^4$ so that $\det[V_T \mid v] = c_T^{\mathsf{T}} v$. Stack these vectors as the rows of a matrix C, and put the 156 normalized projective points in the columns of a matrix D. Each column of CD gives all the determinant values for one point. Replace nonzero values by 1 and zero values by 0, and omit points already used in V; the remaining columns are exactly the Boolean rows $z(V, v)$.

4.3 Compressing the table columns

The input to this step is the collection of Boolean tables $\mathcal{R}(N)$ and $\mathcal{R}(N \setminus f)$ from the preceding subsection. Within each table we drop columns for dependent triples and merge columns for independent triples spanning the same hyperplane. The resulting tables express the compatibility conditions in hyperplane coordinates.

Let M be any rank-preserving matroid extension of N. If a triple T is dependent in N, then $r_M(T \cup \{e\}) \leq 3$, so its bit is zero. If T is independent and $H = \mathrm{cl}_N(T)$, then H is a rank-three hyperplane and

$$\mathrm{cl}_M(T) = \mathrm{cl}_M(H).$$

Indeed, $T \subseteq H \subseteq \mathrm{cl}_M(T)$, which proves both closure inclusions. Hence

$$T \cup \{e\} \in \mathcal{B}(M) \quad \Longleftrightarrow \quad e \notin \mathrm{cl}_M(H).$$

Thus independent triples spanning the same hyperplane have the same bit. Since every hyperplane has a spanning triple, one bit y_H for each rank-three hyperplane H determines all the triple bits of a genuine extension.

Write $\mathcal{H}(N)$ for these hyperplanes. In a table for the deletion of f, the visible hyperplanes are

$$C_f = \{\mathrm{cl}_N(T) : T \in \binom{P \setminus \{f\}}{3},\ r_N(T) = 3\}.$$

The closure of such a triple in $N \setminus f$ is $\mathrm{cl}_N(T) \setminus \{f\}$. Consequently every row of $\mathcal{R}(N \setminus f)$ is constant on triples assigned the same coordinate in C_f, and is zero on dependent triples. Projecting onto one bit per visible hyperplane therefore loses no row information. Let $A_f \subseteq \{0,1\}^{C_f}$ be this projected table, and let $A \subseteq \{0,1\}^{\mathcal{H}(N)}$ be the analogous projection of $\mathcal{R}(N)$.

In these coordinates the search for $\mathcal{S}(N)$ becomes the search for

$$\widehat{\mathcal{S}}(N) = \{y \in \{0,1\}^{\mathcal{H}(N)} : y|_{C_f} \in A_f \text{ for every } f \in P\} \setminus A.$$

To recover the triple bits x, copy y_H to every independent triple spanning H and put zeros on dependent triples. This reverses the compression of each table, so $y|_{C_f} \in A_f$ is equivalent to $\pi_f(x) \in \mathcal{R}(N \setminus f)$, and $y \in A$ is equivalent to $x \in \mathcal{R}(N)$. Conversely, every $x \in \mathcal{S}(N)$ describes a matroid extension by Proposition 3.2, so its triple bits compress to a unique y. The two sets of solutions are therefore in bijection.

Every hyperplane coordinate is seen by at least one deletion: choose $f \notin H$ and a spanning triple of H. That triple avoids f, so $H \in C_f$. The union of the coordinate sets C_f is therefore all of $\mathcal{H}(N)$.

4.4 Searching the compressed tables

Both search methods take the ten compressed deletion tables A_f and the full table A as input. Each A_f is a Boolean *relation* on C_f, meaning a table of allowed rows whose columns are named by the hyperplanes in C_f. The goal is to find all full rows that restrict to an allowed row in every A_f, then exclude the full rows in A. The join combines partial rows directly, while the SAT method encodes their allowed values as clauses on the same variables y_H.

Natural join. Take two tables and compare their rows on the columns they share. For each pair that agrees on every shared column, merge the two rows, keeping one copy of each shared entry. The set of merged rows is the *natural join*. The following two tables share column H_2.

H_1	H_2
0	0
1	1

H_2	H_3
0	1
1	0

The rows with $H_2 = 0$ merge to $(0, 0, 1)$ and those with $H_2 = 1$ merge to $(1, 1, 0)$, both in column order (H_1, H_2, H_3).

Start with one deletion table and join in each remaining table. At every stage a row is an assignment to the columns seen so far that satisfies all the deletions already processed. Adding the next table preserves this statement in both directions. Every merged row satisfies the old tables and the new one. Conversely, any assignment satisfying them restricts to an old row and a new row that agree on their shared columns, so the join produces it. After the last join the columns cover $\mathcal{H}(N)$. Removing the rows in A gives exactly $\widehat{\mathcal{S}}(N)$.

The program groups rows by their bits on the shared columns and chooses the next table to minimize the size of the intermediate join. Once every column has appeared, the program can remove rows in A immediately. Any remaining join can only filter those complete assignments.

SAT enumeration. Use one Boolean variable y_H for each column of the full assignment. To encode membership in A_f, build a binary decision diagram by splitting its rows on successive columns in C_f. A branch with no allowed rows rejects the assignment. A branch that reaches the end with an allowed row accepts it. Share repeated subdiagrams to reduce the encoding.

Give each decision node a Boolean value v for whether it accepts the assignment. At a node testing y_H, let l and h be the values of the branches for 0 and 1. The condition $v \leftrightarrow ((\neg y_H \wedge l) \vee (y_H \wedge h))$ is encoded by

$$\neg v \vee y_H \vee l, \qquad \neg v \vee \neg y_H \vee h,$$
$$v \vee y_H \vee \neg l, \qquad v \vee \neg y_H \vee \neg h.$$

Requiring each table's root to be true enforces $y|_{C_f} \in A_f$ for every f. For each full row $a \in A$, add the clause $\bigvee_{H \in \mathcal{H}(N)} (y_H \neq a_H)$, which requires at least one bit to differ from a. The assignments to the y_H variables that satisfy the resulting formula are exactly $\widehat{\mathcal{S}}(N)$.

Each time the solver finds an assignment to the y_H variables, record it and add a clause of the same form to exclude it from later searches. When the solver returns UNSAT, every solution has been recorded. A search stopped by a resource limit returns `UNKNOWN` and does not establish exhaustion.

The eleven-element classification uses the natural join. The SAT cross-checks in Section 6 compare the solution sets obtained from the same compressed tables. Expanding the resulting rows gives $\mathcal{S}(N)$ and hence the extensions in $\mathcal{D}(N)$, which still require contraction tests.

5 Step 3: testing contractions and certifying the list

Proposition 2.2 tells us how to finish the classification once the sets $\mathcal{D}(N)$ have been recovered. We must test every contraction and then pass from labelled matroids to isomorphism classes. We also verify each final object by certificates that can be checked without rerunning the search.

5.1 The remaining minor conditions

For a simple rank-four matroid M, every contraction M/a has rank three. It may have parallel elements. Its *simplification* $\mathrm{si}(M/a)$ deletes loops and keeps one element of each parallel class.

Lemma 5.1 (The contraction test)**.** *For $M \in \mathcal{D}(N)$, the matroid M is an excluded minor if and only if every $\mathrm{si}(M/a)$ is a quinary rank-three matroid.*

Proof. A matroid is quinary if and only if its simplification is. One direction follows because a simplification is a restriction; for the other, extend a representation of the simplification by zero columns for loops and scalar copies of columns for parallel elements. Thus the stated condition is equivalent to all contractions of M being quinary. Membership in $\mathcal{D}(N)$ already supplies nonrepresentability of M and representability of all its deletions. The immediate-minor criterion from Section 1 now gives the result. □

This is where the rank-three represented catalogue from Step 1 is used. Each simplification has at most ten elements. A simple rank-three matroid on three elements is free and is represented by I_3. At sizes four through ten, exact isomorphism comparison with the corresponding complete quinary catalogue decides the test. In particular, this step needs no assumption about a classification of rank-three excluded minors of arbitrary size.

After applying this test, Proposition 2.2 shows that the retained matroids are exactly the required excluded minors, with possible repetitions from their different parents or labellings. Exact matroid isomorphism removes these repetitions. The final distinctness check also uses colored incidence graphs: one color consists of element vertices, the other of vertices for all bases or, when fewer, all nonbases, with edges given by membership. A color-preserving graph isomorphism induces exactly a matroid isomorphism, since either subset family determines the bases. Canonical forms of these graphs therefore give an exact second check of pairwise nonisomorphism.

5.2 Certifying that every listed object is an excluded minor

The construction proves coverage of the target class. For the opposite inclusion, each final basis family is validated and given two kinds of evidence: representations of all its immediate minors, and a proof that the matroid itself has no representation.

The positive evidence is direct. For each $a \in E(M)$ we save matrices for $M \setminus a$ and M/a, in the order of their remaining labels. An independent modular Gaussian-elimination routine computes the basis family of each matrix and compares it with that of the required minor. It checks every subset of the appropriate size, so agreement establishes equality of the represented matroid and the specified minor. Sage performs a second matrix check. These witnesses certify all the proper-minor conditions using only finite-field arithmetic.

Nonrepresentability requires a different form of evidence. We turn existence of a representation into a finite propositional formula Φ_M and check a certificate that this formula is unsatisfiable. Here is the mathematical condition encoded by the formula. Choose a basis $B = (b_1, \ldots, b_r)$ of M and reduce its columns to the identity. The remaining entries form a

block Z, and fundamental exchanges prescribe its support:

$$Z_{ij} \neq 0 \quad \Longleftrightarrow \quad (B \setminus \{b_i\}) \cup \{j\} \in \mathcal{B}(M), \qquad j \notin B.$$

The unknown nonzero entries range over $\{1, 2, 3, 4\}$. For every r-subset S, the determinant of the columns on S must be nonzero exactly when $S \in \mathcal{B}(M)$. These conditions impose the entire basis family.

Proposition 5.2 (Meaning of the nonrepresentability certificate)**.** *The exact finite-field encoding Φ_M of these determinant conditions is satisfiable if and only if M is quinary.*

Proof. Any representation of M can be reduced on B to identity form. Its remaining entries have the prescribed support and satisfy every determinant condition, so they give a solution. Conversely, a solution supplies a full-rank matrix whose nonsingular r-column submatrices are exactly $\mathcal{B}(M)$; this matrix represents M. The encoding uses exact addition and multiplication tables modulo five, so its satisfying assignments are precisely the solutions of these field conditions. The scaling normalization is justified by Lemma 4.2, and the arithmetic clauses are given in Appendix A. □

For each final matroid, the SAT solver produces a DRAT refutation of Φ_M. DRAT-trim checks this refutation against the saved formula [5]. By Proposition 5.2, a successful check proves that M is nonquinary. Together with the 22 minor matrices, it certifies that M is an excluded minor. These certificates use the final basis records directly; they do not need the deletion tables or their enumeration results.

At this point the logical roles are distinct. Complete parent generation and exhaustive recovery show that every excluded minor reaches the final list. Certificates show that every entry of that list is an excluded minor. Exact isomorphism tests determine how many different classes the list represents. We now give the computational facts that establish these statements for the present rank and size.

6 The computation and the proof of the theorem

The completed run of 27 September 2026 records the outcomes in Table 1. Each row corresponds to a specified mathematical set or operation in the preceding sections.

Stage	Objects counted	Number
Generation	Simple rank-four quinary ten-element parent classes	58,756
Step 1	Connected parent classes in $\mathcal{P}$	58,672
Step 2	Labelled extensions, counted as pairs (N, M) with $M \in \mathcal{D}(N)$	659
Step 3	Classes after all contraction tests and isomorphism deduplication	35

Table 1: The successive sets in the exhaustive classification.

6.1 Evidence that the construction is complete

The represented rank-four generation retains 64,508 field-isomorphism classes at size ten. Abstract deduplication gives the 58,756 simple parent classes in the first row. Exactly 84 are disconnected and are discarded by Lemma 2.1; the remaining 58,672 form $\mathcal{P}$. Aligning the represented classes and closing their automorphism orbits yields 71,547 labelled projective

classes over the full ten-element parent catalogue. The complete nine-element labelled sets supply the other extension tables. All generated levels are listed in Appendix C.

Every one of the 58,672 connected-parent problems finishes by exhaustive natural join, with no unresolved case. The merged logs account for every parent index exactly once, with each parent either excluded for being disconnected or having its finite problem exhausted. The shards have matching input and critical-source fingerprints. By Propositions 4.3 and 3.2, this yields exactly the pairs (N, M) with $M \in \mathcal{D}(N)$: there are 659. This is a count of parent-labelled extensions; it is not yet a count of isomorphism classes of excluded minors.

Testing all contractions using the complete rank-three catalogues, then deduplicating by isomorphism, leaves 35 classes. The independent incidence-graph check confirms that these 35 are pairwise nonisomorphic. For each of them the separate determinant formula has a checked DRAT refutation, and all 22 saved minor matrices pass both modular Gaussian and Sage verification. Thus there are 35 checked refutations and $35 \cdot 22 = 770$ checked matrices.

Proof of Theorem 1.1. To show that there are at most 35 classes, take any rank-four, eleven-element quinary excluded minor M. By Lemma 2.1, a deletion of M is isomorphic to a parent $N \in \mathcal{P}$. The represented generation is complete by Proposition 2.3. After aligning the labels, the recovery theorem, Proposition 3.2, places M among the solutions of the search for N. Its tables are complete by Proposition 4.3, and the recorded search exhausts them. Since every contraction of M is quinary, the complete contraction test of Lemma 5.1 retains it. Hence its class is among the 35 final classes.

To show that there are at least 35 classes, take the 35 final representatives. The checked refutation for each proves its nonrepresentability by Proposition 5.2. Its 22 checked matrices prove representability of every immediate minor, so it is an excluded minor. The independent isomorphism check proves that the representatives belong to 35 different classes. Combining the two bounds proves the classification. □

6.2 Checks on the computation and consequences

The same procedure reproduces the published rank-four catalogues at sizes seven through ten: 5, 92, 219, and 130 classes, respectively [1]. Each comparison is by exact matroid isomorphism with the published archive. At sizes seven and eight the deletion-compatibility inequality $n > 2r$ does not hold, so those calibration runs impose basis exchange explicitly. The rank-four nine-element target does satisfy the inequality.

A separate SAT enumeration of the finite table conditions agrees with the join on every connected-parent problem for the nine-element target. At the ten-element target, the complete sets of 5,991 labelled deletion-minimal candidates agree. These checks test the implementation; the proof of coverage is the parent reduction and recovery argument above. The two enumeration methods share their table construction, and the eleven-element target is exhausted by the join method.

Duality gives a further classification immediately.

Corollary 6.1. *There are exactly 35 isomorphism classes of rank-seven, eleven-element excluded minors for representability over* $\mathbb{F}_5$.

Proof. Field representability is closed under duality, and duality interchanges deletion and contraction. It therefore maps excluded minors to excluded minors and gives a bijection between the rank-four and rank-seven classes on eleven elements. □

Every matroid in the rank-four catalogue has a triangle, so none is paving. This is a property of the resulting classification; no paving restriction is used in the search. Basis and triangle counts accompany the exact descriptions in Appendix B.

A The determinant encoding for nonrepresentability

For the formula Φ_M in Proposition 5.2, choose an ordered basis $B = (b_1, \ldots, b_r)$ and use the normal form of Section 4.1 to write $[I_r \mid Z]$. The support is fixed by the input matroid. Forest entries may be set to 1 by Lemma 4.2, so only the other nonzero entries need variables, each with domain $\{1, 2, 3, 4\}$.

Let S be an r-subset of the ground set and put

$$I(S) = \{i : b_i \notin S\}, \qquad J(S) = S \setminus B.$$

These sets have the same size. Expansion on the identity columns shows that the full determinant on S is, up to sign, $\det Z[I(S), J(S)]$, with the empty determinant equal to 1. It is therefore enough to impose

$$\det Z[I(S), J(S)] \neq 0 \quad \Longleftrightarrow \quad S \in \mathcal{B}(M) \qquad \text{for every } S \in \binom{E(M)}{r}.$$

Every maximal minor is constrained, so these conditions determine the full labelled matroid.

Here is an explicit finite propositional encoding of the field arithmetic. For an expression u with allowed values $D \subseteq \mathbb{F}_5$, introduce indicators $p_{u,a}$ for $a \in D$, requiring exactly one to be true. If $w = u + v$, add the clauses

$$\neg p_{u,a} \ \vee \ \neg p_{v,b} \ \vee \ p_{w,a+b}$$

for every allowed pair a, b, where the sum is taken modulo five. Use the corresponding clauses with ab for multiplication. The output indicators also satisfy the exactly-one condition. Thus selected input values force precisely the correct output value. Constants are substituted directly. A cached Laplace expansion evaluates the determinants, and zero or nonzero requirements constrain their indicators. Induction over the arithmetic expressions proves that the formula represents exactly the field conditions used in Proposition 5.2.

For each listed matroid the saved evidence consists of its basis record, the chosen basis and normalization, the DIMACS formula, a DRAT refutation, and the proof-checker's output. The successful refutation certifies the supplied formula's unsatisfiability; the equivalence with representability is supplied by the mathematical encoding above.

B Exact catalogue

All representatives have ground set $E = \{0, 1, \ldots, 10\}$ and rank four. Each M_i is specified by the listed family $\mathcal{H}_i$ of hyperplanes of size at least four, according to the rule

$$\mathcal{B}(M_i) = \{S \in \binom{E}{4} : S \not\subseteq H \text{ for every } H \in \mathcal{H}_i\}.$$

Thus a four-set is a nonbasis exactly when it is contained in a listed hyperplane. This specifies the whole basis family: every dependent four-set of a rank-four matroid lies in a hyperplane, while hyperplanes of size three contain no four-set and may be omitted. The descriptions have been checked by reconstructing the basis records from them.

The notation M_i abbreviates the stable identifier `GF5-r4-n11-` followed by i in four digits. For example, M_1 is `GF5-r4-n11-0001`. Proof directories are zero-indexed, so its certificates are in `matroid-0000`. Basis and triangle counts identify the corresponding records in the machine-readable catalogue. Across the catalogue there are between 210 and 262 bases, and the triangle distribution is

triangles	4	5	6	7	8	10	12	13
classes	3	5	5	10	9	1	1	1

M_1: **250 bases; 6 triangles.** $\{0,1,2,5,7,9\}$, $\{0,1,4,10\}$, $\{0,2,3,4,8\}$, $\{0,2,6,10\}$, $\{0,3,5,10\}$, $\{0,3,6,9\}$, $\{0,4,5,6\}$, $\{0,6,7,8\}$, $\{0,8,9,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8\}$, $\{1,2,5,10\}$, $\{1,3,7,9\}$, $\{1,4,7,9\}$, $\{1,6,7,8,9,10\}$, $\{2,3,4,7,10\}$, $\{2,3,4,9\}$, $\{2,6,7,8\}$, $\{2,8,9,10\}$, $\{3,6,7,8\}$, $\{3,8,9,10\}$, $\{4,5,6,7,8\}$, $\{4,5,6,9\}$, $\{4,5,6,10\}$, $\{4,8,9,10\}$, $\{5,8,9,10\}$.

M_2: **233 bases; 7 triangles.** $\{0,1,2,6\}$, $\{0,1,4,7\}$, $\{0,1,8,9,10\}$, $\{0,2,3,4,5\}$, $\{0,2,7,9\}$, $\{0,3,6,9\}$, $\{0,3,7,10\}$, $\{0,4,6,8\}$, $\{0,5,6,10\}$, $\{0,5,7,8\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,6,8\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,4,7,8\}$, $\{1,4,7,9\}$, $\{1,4,7,10\}$, $\{2,3,4,5,8,9,10\}$, $\{6,7,8,9,10\}$.

M_3: **252 bases; 7 triangles.** $\{0,1,2,5,7\}$, $\{0,1,3,6\}$, $\{0,1,4,9,10\}$, $\{0,2,3,4,8\}$, $\{0,2,6,9\}$, $\{0,3,5,7\}$, $\{0,4,5,6,7\}$, $\{0,5,7,8,9\}$, $\{0,5,7,10\}$, $\{0,6,8,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8\}$, $\{1,2,5,9\}$, $\{1,2,5,10\}$, $\{1,3,6,7\}$, $\{1,3,6,8,10\}$, $\{1,3,6,9\}$, $\{1,7,8,9\}$, $\{2,3,4,7,10\}$, $\{2,3,4,9\}$, $\{2,6,8,10\}$, $\{2,7,8,9\}$, $\{3,5,9,10\}$, $\{3,7,8,9\}$, $\{4,5,6,8,10\}$, $\{4,5,6,9\}$, $\{4,7,8,9\}$, $\{6,7,8,9,10\}$.

M_4: **250 bases; 5 triangles.** $\{0,1,2,5,7\}$, $\{0,1,3,10\}$, $\{0,1,4,8\}$, $\{0,2,3,4,8,9\}$, $\{0,2,6,10\}$, $\{0,3,5,7\}$, $\{0,4,5,7,8\}$, $\{0,4,6,8\}$, $\{0,4,8,10\}$, $\{0,5,6,7,9\}$, $\{0,5,7,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8\}$, $\{1,2,5,9\}$, $\{1,2,5,10\}$, $\{1,6,7,8,9,10\}$, $\{2,3,4,7,10\}$, $\{2,6,7,9\}$, $\{3,5,8,10\}$, $\{3,6,7,9\}$, $\{4,5,9,10\}$, $\{4,6,7,9\}$.

M_5: **247 bases; 7 triangles.** $\{0,1,2,6,8\}$, $\{0,1,7,9\}$, $\{0,2,3,4,5,7\}$, $\{0,3,6,8,9,10\}$, $\{0,4,6,8\}$, $\{0,5,6,8\}$, $\{0,6,7,8\}$, $\{1,2,3,4,5,6\}$, $\{1,2,6,7\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,3,7,8\}$, $\{1,4,7,10\}$, $\{1,5,8,9,10\}$, $\{2,3,4,5,8\}$, $\{2,3,4,5,9\}$, $\{2,3,4,5,10\}$, $\{2,7,8,9,10\}$, $\{4,6,7,9\}$, $\{4,8,9,10\}$, $\{5,6,7,10\}$.

M_6: **241 bases; 8 triangles.** $\{0,1,2,7,9,10\}$, $\{0,1,6,8\}$, $\{0,2,3,4,5,8\}$, $\{0,3,7,9,10\}$, $\{0,4,7,9,10\}$, $\{0,5,6,7,9,10\}$, $\{0,7,8,9,10\}$, $\{1,2,3,4,5,6\}$, $\{1,3,8,9\}$, $\{1,4,8,10\}$, $\{1,5,7,8\}$, $\{2,3,4,5,7\}$, $\{2,3,4,5,9\}$, $\{2,3,4,5,10\}$, $\{2,6,8,10\}$, $\{3,6,7,8\}$, $\{4,6,8,9\}$.

M_7: **249 bases; 7 triangles.** $\{0,1,2,6,8\}$, $\{0,1,7,9,10\}$, $\{0,2,3,4,5,7\}$, $\{0,3,6,8,9\}$, $\{0,4,6,8\}$, $\{0,5,6,8,10\}$, $\{0,6,7,8\}$, $\{1,2,3,4,5,6\}$, $\{1,2,6,7\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,3,7,8\}$, $\{1,4,8,10\}$, $\{1,5,8,9\}$, $\{2,3,4,5,8\}$, $\{2,3,4,5,9\}$, $\{2,3,4,5,10\}$, $\{2,7,8,9,10\}$, $\{3,7,9,10\}$, $\{4,6,7,9,10\}$, $\{5,7,9,10\}$.

M_8: **252 bases; 7 triangles.** $\{0,1,2,5,7\}$, $\{0,1,3,6\}$, $\{0,1,8,9\}$, $\{0,2,3,4,8\}$, $\{0,3,5,7\}$, $\{0,4,5,6,7\}$, $\{0,5,7,8\}$, $\{0,5,7,9,10\}$, $\{0,6,8,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8,9\}$, $\{1,2,5,10\}$, $\{1,3,6,7\}$, $\{1,3,6,8,9\}$, $\{1,3,6,10\}$, $\{1,4,8,9\}$, $\{1,7,8,9,10\}$, $\{2,3,4,7,9,10\}$, $\{4,5,6,8\}$, $\{4,5,6,9\}$, $\{4,5,6,10\}$, $\{6,7,9,10\}$.

M_9: **255 bases; 6 triangles.** $\{0,1,2,5,7\}$, $\{0,1,3,8\}$, $\{0,1,6,9,10\}$, $\{0,2,3,4,8\}$, $\{0,2,6,9\}$, $\{0,3,5,8\}$, $\{0,3,6,8,9\}$, $\{0,3,7,8,10\}$, $\{0,4,5,6,9\}$, $\{0,6,7,9\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8,9\}$, $\{1,2,5,10\}$, $\{1,3,7,9\}$, $\{1,7,8,10\}$, $\{2,3,4,7\}$, $\{2,3,4,9\}$, $\{2,3,4,10\}$, $\{2,6,7,8,10\}$, $\{3,5,9,10\}$, $\{4,5,6,7\}$, $\{4,5,6,8\}$, $\{4,5,6,10\}$, $\{4,7,8,9,10\}$, $\{5,7,8,10\}$.

M_{10}: **239 bases; 8 triangles.** $\{0,1,2,6,8\}$, $\{0,1,4,9,10\}$, $\{0,1,5,7\}$, $\{0,2,3,4,5\}$, $\{0,2,7,9\}$, $\{0,3,6,8,9\}$, $\{0,3,7,10\}$, $\{0,4,6,7,8\}$, $\{0,5,6,8\}$, $\{0,6,8,10\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,5,7,8\}$, $\{1,5,7,9\}$, $\{1,5,7,10\}$, $\{2,3,4,5,8,10\}$, $\{2,3,4,5,9\}$, $\{4,6,7,9\}$, $\{4,6,7,10\}$, $\{5,6,9,10\}$, $\{7,8,9,10\}$.

M_{11}: **212 bases; 10 triangles.** $\{0,1,2,6,8,9\}$, $\{0,1,4,10\}$, $\{0,1,5,7\}$, $\{0,2,3,4,5\}$, $\{0,2,7,10\}$, $\{0,3,6,10\}$, $\{0,3,7,9\}$, $\{0,4,6,7\}$, $\{0,5,8,10\}$, $\{1,2,3,4,5,6,7,8\}$, $\{1,2,6,8,10\}$, $\{1,5,7,9\}$, $\{1,5,7,10\}$, $\{2,3,4,5,9,10\}$, $\{4,6,7,9\}$, $\{4,6,7,10\}$, $\{7,8,9,10\}$.

M_{12}: **242 bases; 8 triangles.** $\{0,1,2,6,8\}$, $\{0,1,3,10\}$, $\{0,1,4,9\}$, $\{0,1,5,7\}$, $\{0,2,3,4,5,9\}$, $\{0,4,6,7,9\}$, $\{0,4,8,9\}$, $\{0,4,9,10\}$, $\{0,5,8,10\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,6,9,10\}$, $\{1,5,7,8\}$, $\{1,5,7,9\}$, $\{1,5,7,10\}$, $\{2,3,4,5,8\}$, $\{2,3,4,5,10\}$, $\{3,7,8,9,10\}$, $\{4,6,7,8\}$, $\{4,6,7,10\}$.

M_{13}: **254 bases; 5 triangles.** $\{0,1,2,5,7\}$, $\{0,1,4,8\}$, $\{0,2,3,4,8,9\}$, $\{0,3,5,7\}$, $\{0,4,5,7,8\}$, $\{0,4,6,8\}$, $\{0,4,8,10\}$, $\{0,5,6,7,9,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8\}$, $\{1,2,5,9\}$, $\{1,2,5,10\}$, $\{1,3,7,8\}$, $\{1,6,8,9,10\}$, $\{2,3,4,7,10\}$, $\{2,6,9,10\}$, $\{3,6,9,10\}$, $\{4,6,9,10\}$.

M_{14}: **257 bases; 4 triangles.** $\{0,1,2,5,7,9\}$, $\{0,1,4,10\}$, $\{0,2,3,4,8\}$, $\{0,3,6,7,9,10\}$, $\{0,4,7,9\}$, $\{0,5,6,8\}$, $\{0,7,8,9\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8\}$, $\{1,2,5,10\}$, $\{1,3,7,8\}$, $\{1,6,8,9,10\}$, $\{2,3,4,7\}$, $\{2,3,4,9\}$, $\{2,3,4,10\}$, $\{2,6,9,10\}$, $\{4,5,7,8,10\}$, $\{4,6,9,10\}$, $\{5,6,9,10\}$.

M_{15}: **262 bases; 4 triangles.** $\{0,1,2,5,7\}$, $\{0,1,4,10\}$, $\{0,2,3,4,8\}$, $\{0,3,5,7,10\}$, $\{0,4,5,7\}$, $\{0,5,6,7,8,9\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8\}$, $\{1,2,5,9\}$, $\{1,2,5,10\}$, $\{1,3,7,9\}$, $\{1,4,7,8\}$, $\{1,6,8,9,10\}$, $\{2,3,4,7\}$, $\{2,3,4,9,10\}$, $\{2,6,8,9\}$, $\{2,7,8,10\}$, $\{3,6,8,9\}$, $\{4,6,7,10\}$, $\{4,6,8,9\}$.

M_{16}: **210 bases; 13 triangles.** $\{0,1,2,7\}$, $\{0,1,4,9,10\}$, $\{0,1,5,8\}$, $\{0,2,3,4,5,6\}$, $\{0,2,8,9\}$, $\{0,3,7,9\}$, $\{0,3,8,10\}$, $\{0,4,7,8\}$, $\{0,6,7,10\}$, $\{1,2,3,4,5,6,7,8\}$, $\{1,2,7,9\}$, $\{1,2,7,10\}$, $\{1,5,8,9\}$, $\{1,5,8,10\}$, $\{2,3,4,5,6,9\}$, $\{2,3,4,5,6,10\}$, $\{4,7,8,9\}$, $\{4,7,8,10\}$, $\{5,7,9,10\}$, $\{6,8,9,10\}$.

M_{17}: **245 bases; 7 triangles.** $\{0,1,2,5,7,9\}$, $\{0,1,4,8\}$, $\{0,2,3,4,8,9\}$, $\{0,2,6,9\}$, $\{0,2,9,10\}$, $\{0,3,5,7\}$, $\{0,4,5,7,8\}$, $\{0,4,6,8\}$, $\{0,4,8,10\}$, $\{0,5,6,7,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8,10\}$, $\{1,3,7,8,9\}$, $\{1,4,7,9\}$, $\{1,6,7,9\}$, $\{1,7,9,10\}$, $\{2,3,4,7,10\}$, $\{3,5,8,9\}$, $\{3,6,8,9,10\}$.

M_{18}: **242 bases; 6 triangles.** $\{0,1,2,5,8,10\}$, $\{0,1,3,10\}$, $\{0,1,4,9,10\}$, $\{0,1,6,10\}$, $\{0,1,7,10\}$, $\{0,2,3,4,9\}$, $\{0,3,5,7\}$, $\{0,4,5,6,9\}$, $\{0,4,7,9\}$, $\{0,4,8,9\}$, $\{0,6,7,8\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,5,9\}$, $\{1,7,8,9\}$, $\{2,3,4,8\}$, $\{2,3,4,10\}$, $\{2,7,9,10\}$, $\{3,5,7,8\}$, $\{3,5,7,9\}$, $\{3,5,7,10\}$, $\{3,6,8,9,10\}$, $\{4,5,6,8\}$, $\{4,5,6,10\}$, $\{4,7,8,10\}$.

M_{19}: **244 bases; 8 triangles.** $\{0,1,2,6,7,9\}$, $\{0,1,3,9\}$, $\{0,1,4,8,9\}$, $\{0,1,5,9,10\}$, $\{0,2,3,4,5,7\}$, $\{0,2,7,8\}$, $\{0,2,7,10\}$, $\{0,6,8,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,6,8\}$, $\{1,2,6,10\}$, $\{1,3,8,10\}$, $\{2,3,4,5,8\}$, $\{2,3,4,5,9\}$, $\{2,3,4,5,10\}$, $\{2,8,9,10\}$, $\{3,6,7,8,9\}$, $\{4,6,7,9,10\}$, $\{5,6,7,9\}$, $\{5,7,8,10\}$.

M_{20}: **256 bases; 6 triangles.** $\{0,1,2,5,7\}$, $\{0,1,3,9,10\}$, $\{0,1,4,6\}$, $\{0,2,3,4,6\}$, $\{0,3,5,7\}$, $\{0,4,5,6,7\}$, $\{0,4,6,8,10\}$, $\{0,4,6,9\}$, $\{0,5,7,8,9\}$, $\{0,5,7,10\}$, $\{1,2,3,4,5\}$, $\{1,2,5,6\}$, $\{1,2,5,8,9\}$, $\{1,2,5,10\}$, $\{1,3,6,7,8\}$, $\{1,4,7,9\}$, $\{1,4,8,10\}$, $\{2,3,4,7\}$, $\{2,3,4,8,10\}$, $\{2,3,4,9\}$, $\{2,6,7,9,10\}$, $\{3,5,6,10\}$, $\{3,5,8,9\}$, $\{4,5,8,9,10\}$, $\{4,7,8,10\}$, $\{5,6,8,9\}$.

M_{21}: **226 bases; 8 triangles.** $\{0,1,2,6,8,9\}$, $\{0,1,4,7\}$, $\{0,1,5,10\}$, $\{0,2,3,4,5\}$, $\{0,2,7,10\}$, $\{0,3,7,9\}$, $\{0,3,8,10\}$, $\{0,4,6,10\}$, $\{0,5,7,8\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,6,10\}$, $\{1,3,8,10\}$, $\{1,4,7,8\}$, $\{1,4,7,9\}$, $\{1,4,7,10\}$, $\{2,3,4,5,8,9,10\}$, $\{2,7,8,9\}$, $\{3,6,8,10\}$, $\{3,7,8,10\}$, $\{6,7,9,10\}$.

M_{22}: **261 bases; 4 triangles.** $\{0,1,2,5,7\}$, $\{0,1,6,8,9\}$, $\{0,2,3,4,8\}$, $\{0,2,6,9\}$, $\{0,3,5,7\}$, $\{0,3,6,9\}$, $\{0,4,5,7\}$, $\{0,4,6,9\}$, $\{0,5,6,7,9,10\}$, $\{0,5,7,8\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8,10\}$, $\{1,2,5,9\}$, $\{1,3,7,10\}$, $\{1,4,7,9\}$, $\{2,3,4,7\}$, $\{2,3,4,9,10\}$, $\{2,7,8,9\}$, $\{3,5,8,9\}$, $\{3,6,7,8\}$, $\{4,7,8,10\}$.

M_{23}: **220 bases; 8 triangles.** $\{0,1,2,6\}$, $\{0,1,4,7\}$, $\{0,1,5,9\}$, $\{0,1,8,10\}$, $\{0,2,3,4,5,10\}$, $\{0,2,8,9\}$, $\{0,3,6,8\}$, $\{0,5,7,8\}$, $\{0,6,7,9,10\}$, $\{1,2,3,4,5,6,7,8\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,3,9,10\}$, $\{1,4,7,9\}$, $\{1,4,7,10\}$, $\{2,3,4,5,9\}$, $\{3,6,8,9\}$, $\{3,6,8,10\}$, $\{4,8,9,10\}$, $\{5,7,8,9\}$, $\{5,7,8,10\}$.

M_{24}: **220 bases; 8 triangles.** $\{0,1,2,6\}$, $\{0,1,4,7\}$, $\{0,1,8,9,10\}$, $\{0,2,3,4,5\}$, $\{0,2,7,9\}$, $\{0,3,6,9\}$, $\{0,3,7,10\}$, $\{0,4,6,8\}$, $\{0,5,6,10\}$, $\{0,5,7,8\}$, $\{1,2,3,4,5,6,7,8\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,4,7,9\}$, $\{1,4,7,10\}$, $\{2,3,4,5,9,10\}$, $\{4,6,8,9\}$, $\{4,6,8,10\}$, $\{5,7,8,9\}$, $\{5,7,8,10\}$, $\{6,7,9,10\}$.

M_{25}: **243 bases; 7 triangles.** $\{0,1,2,6\}$, $\{0,1,4,7\}$, $\{0,1,8,9,10\}$, $\{0,2,3,4,5\}$, $\{0,2,7,10\}$, $\{0,2,8,9\}$, $\{0,3,6,10\}$, $\{0,3,8,9\}$, $\{0,4,6,8,9\}$, $\{0,5,7,8,9\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,6,8\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,4,7,8\}$, $\{1,4,7,9\}$, $\{1,4,7,10\}$, $\{2,3,4,5,8\}$, $\{2,3,4,5,9,10\}$, $\{3,7,8,10\}$, $\{5,6,8,10\}$, $\{6,7,9,10\}$.

M_{26}: **243 bases; 7 triangles.** $\{0,1,2,6\}$, $\{0,1,4,7\}$, $\{0,1,8,9,10\}$, $\{0,2,3,4,5\}$, $\{0,2,8,9\}$, $\{0,3,6,8,9\}$, $\{0,3,7,10\}$, $\{0,4,6,10\}$, $\{0,4,8,9\}$, $\{0,5,7,8,9\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,6,8\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,4,7,8\}$, $\{1,4,7,9\}$, $\{1,4,7,10\}$, $\{2,3,4,5,8\}$, $\{2,3,4,5,9,10\}$, $\{2,7,8,10\}$, $\{5,6,8,10\}$, $\{6,7,9,10\}$.

M_{27}: **228 bases; 12 triangles.** $\{0,1,2,7\}$, $\{0,1,4,9\}$, $\{0,1,5,8,10\}$, $\{0,2,3,4,5,6\}$, $\{0,2,8,9,10\}$, $\{0,3,7,9\}$, $\{0,3,8,10\}$, $\{0,4,7,8,10\}$, $\{0,6,8,10\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,7,8\}$, $\{1,2,7,9\}$, $\{1,2,7,10\}$, $\{1,3,9,10\}$, $\{1,6,8,9\}$, $\{2,3,4,5,6,8\}$, $\{2,3,4,5,6,9\}$, $\{2,3,4,5,6,10\}$, $\{5,7,8,9\}$, $\{6,7,9,10\}$.

M_{28}: **248 bases; 5 triangles.** $\{0,1,2,5,8\}$, $\{0,1,7,9\}$, $\{0,2,3,4\}$, $\{0,2,7,9\}$, $\{0,3,6,7,9,10\}$, $\{0,4,5,6\}$, $\{0,4,7,8,9\}$, $\{0,5,7,9\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,5,9,10\}$, $\{1,3,8,10\}$, $\{2,3,4,8\}$, $\{2,3,4,9\}$, $\{2,3,4,10\}$, $\{2,6,8,9\}$, $\{3,6,7,8\}$, $\{4,5,6,8,10\}$, $\{4,5,6,9\}$.

M_{29}: **216 bases; 8 triangles.** $\{0,1,2,9\}$, $\{0,1,6,7,8,10\}$, $\{0,2,3,4,5,10\}$, $\{0,3,8,9\}$, $\{0,4,7,9\}$, $\{0,5,6,9\}$, $\{1,2,3,4,5,6,7,8\}$, $\{1,3,9,10\}$, $\{1,6,7,8,9\}$, $\{2,3,4,5,9\}$, $\{2,7,9,10\}$, $\{4,6,9,10\}$, $\{5,8,9,10\}$.

M_{30}: **256 bases; 5 triangles.** $\{0,1,2,5,7\}$, $\{0,1,4,9,10\}$, $\{0,2,3,4,8\}$, $\{0,2,6,9,10\}$, $\{0,3,5,9,10\}$, $\{0,4,5,6\}$, $\{0,6,7,8\}$, $\{0,7,9,10\}$, $\{0,8,9,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8,9\}$, $\{1,2,5,10\}$, $\{1,6,7,8,10\}$, $\{2,3,4,7,10\}$, $\{2,3,4,9\}$, $\{2,6,7,8\}$, $\{3,6,7,8,9\}$, $\{4,5,6,7,8\}$, $\{4,5,6,9\}$, $\{4,5,6,10\}$.

M_{31}: **239 bases; 7 triangles.** $\{0,1,2,6,8,9\}$, $\{0,1,4,7\}$, $\{0,2,3,4,5,10\}$, $\{0,3,6,8\}$, $\{0,4,6,8\}$, $\{0,5,6,8\}$, $\{0,5,7,9\}$, $\{0,6,7,8,10\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,6,10\}$, $\{1,3,9,10\}$, $\{1,4,7,8\}$, $\{1,4,7,9\}$, $\{1,4,7,10\}$, $\{2,3,4,5,8\}$, $\{2,3,4,5,9\}$, $\{2,7,9,10\}$, $\{3,7,8,9\}$, $\{4,6,9,10\}$, $\{5,8,9,10\}$.

M_{32}: **241 bases; 7 triangles.** $\{0,1,2,5,8\}$, $\{0,1,3,7\}$, $\{0,1,4,9,10\}$, $\{0,2,3,4\}$, $\{0,2,6,7,9\}$, $\{0,3,5,8,9\}$, $\{0,3,6,10\}$, $\{0,4,5,6,8\}$, $\{0,5,7,8\}$, $\{0,5,8,10\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,5,9\}$, $\{1,2,5,10\}$, $\{1,3,7,8,9\}$, $\{1,3,7,10\}$, $\{2,3,4,8,9\}$, $\{2,3,4,10\}$, $\{2,6,7,8,10\}$, $\{3,6,8,9\}$, $\{3,8,9,10\}$, $\{4,5,6,9\}$, $\{4,5,6,10\}$, $\{5,7,9,10\}$.

M_{33}: **241 bases; 6 triangles.** $\{0,1,2,5,8,9\}$, $\{0,1,3,10\}$, $\{0,2,3,4\}$, $\{0,2,6,10\}$, $\{0,3,6,7,9\}$, $\{0,4,5,6\}$, $\{0,4,7,8\}$, $\{0,4,9,10\}$, $\{0,5,7,10\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,5,10\}$, $\{1,3,8,9\}$, $\{1,4,8,9,10\}$, $\{1,6,8,9\}$, $\{1,7,8,9\}$, $\{2,3,4,8\}$, $\{2,3,4,9,10\}$, $\{3,6,7,8,10\}$, $\{4,5,6,8\}$, $\{4,5,6,9,10\}$, $\{4,7,9,10\}$.

M_{34}: **258 bases; 5 triangles.** $\{0,1,2,5,7,9\}$, $\{0,1,4,10\}$, $\{0,2,3,4,8\}$, $\{0,3,5,7\}$, $\{0,3,6,9,10\}$, $\{0,4,5,6,7\}$, $\{0,5,7,8\}$, $\{0,5,7,10\}$, $\{1,2,3,4,5,6\}$, $\{1,2,5,8,10\}$, $\{1,3,7,9\}$, $\{1,4,7,9\}$, $\{1,6,7,8,9\}$, $\{1,7,9,10\}$, $\{2,3,4,7\}$, $\{2,3,4,9\}$, $\{2,3,4,10\}$, $\{2,6,7,10\}$, $\{3,5,8,9\}$, $\{3,7,8,10\}$, $\{4,5,6,8\}$, $\{4,5,6,9\}$, $\{4,5,6,10\}$, $\{4,8,9,10\}$.

M_{35}: **238 bases; 8 triangles.** $\{0,1,2,6,8\}$, $\{0,1,5,7,9\}$, $\{0,2,3,4,5,10\}$, $\{0,3,6,8,9\}$, $\{0,4,6,7,8\}$, $\{0,5,6,8\}$, $\{0,6,8,10\}$, $\{1,2,3,4,5,6,7\}$, $\{1,2,6,9\}$, $\{1,2,6,10\}$, $\{1,3,9,10\}$, $\{1,5,7,8,10\}$, $\{2,3,4,5,8\}$, $\{2,3,4,5,9\}$, $\{2,7,9,10\}$, $\{4,6,7,9\}$, $\{4,6,7,10\}$, $\{4,8,9,10\}$, $\{5,6,9,10\}$.

C Computational records and reproduction

This appendix identifies the saved evidence for each step of the proof. All paths are relative to the repository root. The exact catalogue is `results/r4-n11/excluded.jsonl`, with SHA-256 digest

`7940989ae051b32496dbfd7e4717d405431bcfd89da69d0dd522517552c76441.`

Its equivalent hyperplane descriptions are in `results/r4-n11/hyperplanes.json`.

C.1 Parent coverage and exhaustive search

Table 2 gives the completed represented levels. The rank-four size-four entry is the seed I_4; rank-three generation starts from I_3. The counts distinguish represented classes, which must be retained during generation, from abstract matroid classes used to index the parent searches.

The main scripts correspond to the proof as follows.

Parent generation. `scripts/generate_parents.py` generates all represented rank-four and rank-three levels through size ten. The completed abstract lists index the parents and supply the contraction comparisons.

Labelled representations. `scripts/prepare_representations.py` aligns the represented classes and closes under automorphisms. The normalization and point-extension tables are implemented in `gf5minors/projective.py`.

Size	Rank 3 field	Rank 3 abstract	Rank 4 field	Rank 4 abstract
4	2	2	1	1
5	4	4	3	3
6	11	9	12	11
7	22	18	58	44
8	42	34	460	303
9	92	82	5,001	3,763
10	174	168	64,508	58,756

Table 2: Complete simple represented catalogues. "Field" counts field-isomorphism classes; "abstract" counts matroid classes.

Recovery and contraction testing. `scripts/search_compatible_deletions.py` builds the tables for each parent and tests contractions of the recovered matroids. The exact join is in `gf5minors/table_join.py`; the alternative SAT solver is in `gf5minors/table_sat.py`.

Merging and counting. `scripts/merge_search.py` checks parent coverage and removes isomorphic repetitions. `scripts/final_audit.py` checks certificate coverage and distinctness by canonical incidence graphs.

The four ranges in Table 3 partition $[0, 58756)$. Their logs record either exhaustion or the disconnected-parent exclusion for every index. The merge checks the ranges for gaps and overlaps, terminal statuses, and agreement of input and critical-source fingerprints.

Start	Stop	Parents	Candidates	Seconds
0	14,689	14,689	514	1794.54
14,689	29,378	14,689	106	1629.19
29,378	44,067	14,689	29	1430.24
44,067	58,756	14,689	10	1299.00

Table 3: Exhaustive target-search ranges. Candidate counts precede contraction filtering and global isomorphism deduplication.

The global coverage record is `results/r4-n11/summary.json`. Detailed parent logs and input manifests are in `runs/r4n11/shard-00/` through `runs/r4n11/shard-03/`. The final audit and statistics are `results/r4-n11/verification.json` and `results/r4-n11/statistics.json`. The target search used four Slurm tasks, each requesting one CPU and 4 GB of memory. The longest range took about 30 minutes.

C.2 Certificates and their replay

`scripts/certify_nonrepresentation.py` constructs the separate determinant formulas and checks their DRAT proofs. The 35 proof directories under `results/r4-n11/nonrepresentation/` are named `matroid-0000` through `matroid-0034`. For M_i, the directory index is $i - 1$. Its files `representation.cnf` and `proof.drat` can be replayed with

```
vendor/drat-trim/drat-trim FORMULA PROOF
```

A successful check prints `s VERIFIED`.

`scripts/certify_minors.py` constructs and checks the immediate-minor matrices. All 770 are stored in `results/r4-n11/minors/minor-witnesses.jsonl`. The remaining column labels are in increasing parent order, renumbered to $0, \ldots, 9$. The pure modular checker `gf5minors.matroid.verify_representation` can check these matrices directly from the basis records, independently of the search and the represented catalogues.

The completed computation records Sage 10.9 and, in a separate environment, Python 3.11.16, python-sat 1.9.dev15, and NumPy 2.4.6. Sage supplies represented extensions, isomorphism, and graph operations [4]. The proof checker is pinned to DRAT-trim commit `2e3b2dc0ecf938addbd779d42877b6ed69d9a985`. The computational proof relies on the correctness of parent generation, representation orbits, table construction, exhaustive enumeration, and isomorphism handling. The DRAT checks certify the final determinant formulas; coverage of the classification is supplied by the exhaustive construction.

On the original cluster, run computation through Slurm with the existing `my-sage` and `matroid1` environments. `slurm/research.slurm` selects the environment and caps numerical-library threads at one. Full reproduction commands, in dependency order, are given in `docs/REPRODUCING.md`. Use fresh output directories for new runs. The reproducibility archive `results/r4-n11-reproducibility.tar.gz` and its adjacent JSON manifest preserve the original result and its inputs; that archive predates this exposition.

References

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[3] J. Oxley, *Matroid Theory*, second edition, Oxford Graduate Texts in Mathematics 21, Oxford University Press, 2011.

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